

Density-Calibrated Neural Implicit Field for Ballistic Helical GLAD: Scalar Porosity Prediction, Label-Quality Diagnosis, and the Path to Binary Voxel Occupancy

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Contents

1	Introduction	4
2	Methods	7
2.1	Dataset and voxel representation	7
2.2	FiLM-conditioned NIF architecture (Experiments 2–3)	7
2.3	Experiment 1: full-field density reconstruction (NIF v5)	9
2.4	Experiment 2: soft-label spatial occupancy (PINN-BCE)	9

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14	2.5	Experiment 3: hard-label spatial occupancy	11
15	2.6	Oracle predictor and root-cause diagnostic	11
16	3	Results	12
17	3.1	Experiment 1 — full-field reconstruction at trained angles: strong positive	
18		result, but no angular interpolation	12
19	3.1.1	Held-out angular interpolation: confirmed negative result	13
20	3.2	Experiment 2 — soft-label BCE: confirmed negative result	13
21	3.2.1	Global metrics	15
22	3.2.2	Per-alpha threshold sweep	15
23	3.3	Label-quality diagnosis: the oracle predictor	16
24	3.4	Experiment 3 — hard-label BCE: restoring spatial learning	19
25	3.4.1	Training dynamics	19
26	3.4.2	Per-alpha threshold sweep	19
27	4	Discussion	21
28	4.1	Positive full-field result within a narrow angular window, and its limits . .	21
29	4.2	The soft-label equilibrium trap	22
30	4.3	The oracle as an architecture-independent test	23
31	4.4	Hard labels as a minimum prerequisite	24
32	4.5	Multi-seed pooling at $\alpha = 80^\circ, 85^\circ$	24
33	4.6	Testing the remaining prescriptions: spatial encoding, loss framing, and	
34		partial-realization conditioning	25
35	4.7	Architecture-independent cross-validation: classical reconstruction and global	
36		connectivity agree with the diagnosis	29
37	4.8	Limitations	34
38	5	Conclusion	34

Abstract

A neural implicit field (NIF) is a compact machine-learning model that takes a small set of numbers as input – here, the deposition angle α and a point’s spatial coordinates (x, y, z) – and outputs a prediction, offering a potentially much faster alternative to a full physics simulation for predicting film structure in ballistic helical glancing-angle deposition (GLAD). We report three successive attempts to build such a model, each diagnosing why the previous one fell short. First, a variant conditioned on a reference simulation via feature-wise linear modulation (FiLM) successfully reconstructs the full 3D density field, with mean spatial Pearson correlation $r = 0.984$ – but only for deposition angles within the $75\text{--}87^\circ$ range it was trained on, not beyond it. This figure is also in-sample: evaluated at three angles inside that same range but withheld from training (78.75° , 83.5° , 84.25°), the identical checkpoint collapses to $r \approx 0.003$ mean, showing the network reproduces the seven trained density fields individually rather than learning a function that interpolates smoothly between them, despite an explicit angle-smoothness regulariser applied during training. Second, when instead trained to predict binary voxel occupancy (whether a given point is solid or empty) using softened training labels, the same architecture collapses to the trivial prediction (always guessing the same value) at every angle, with essentially no improvement over that trivial baseline ($\Delta\text{IoU} < 0.003$). An independently reproduced diagnostic test – an (α, z) -oracle check – isolates the cause precisely: this is a failure of how the training labels were constructed, not a limitation of the model architecture or a lack of depth information in the input – 90–98% of the softened labels fall in an ambiguous middle range $(0.1, 0.9)$, which removes the sharp boundary signal the model needs to learn from. Switching to sharp (hard binary) labels restores genuine spatial learning, but only for the most sparsely-packed film in the dataset ($\alpha = 89^\circ$, improvement $\Delta\text{IoU} = +0.029$); denser films remain unlearnable. We then tested three further fixes – a higher-resolution multi-resolution hash-grid input encoding, predicting a continuous density instead of binary occupancy, and conditioning the model on a partial (incomplete) realization of the film (damped and fully unclamped FiLM variants) – and none produced meaningful spatial learning at the angles

where multiple independent simulation runs were available for testing. For the partial-realization approach specifically, we checked directly whether the problem was simply too few independent runs to learn from (three per angle): enlarging one angle’s dataset to eight independent runs and retraining did not help – a statistical permutation test still shows no learned signal for either FiLM variant tested ($p = 0.35$ and $p = 0.53$, across 56 pairs of runs compared), ruling out limited data as the explanation. This diagnosis is further corroborated by four additional, architecture-independent checks – a classical stochastic-reconstruction baseline, an independently-computed global-connectivity descriptor, a spatially-aware convolutional network, and a resolution-preserving local-conditioning variant – which all converge on the same conclusion: the missing information is not resident in any low-order structural descriptor of these films, for any method tested. The result is both a verified, step-by-step diagnostic framework for this class of failure in GLAD voxel-occupancy prediction, and a tested, strengthened negative conclusion: none of the remedies explored here, including simply providing more training data, resolves the underlying non-identifiability – multiple film realizations grown under identical conditions are genuinely indistinguishable from coordinates and angle alone, which is exactly the signal a spatial coordinate-based model would need in order to learn a meaningful occupancy boundary.

Keywords: glancing-angle deposition; neural implicit field; FiLM conditioning; voxel occupancy; label quality; binary cross-entropy; ballistic simulation; porosity prediction

1 Introduction

Glancing-angle deposition (GLAD) produces self-assembled nanostructures whose morphology is exquisitely sensitive to the deposition angle α , an angle-controlled columnar growth process reviewed extensively [Hawkeye et al., 2014, Barranco et al., 2016, Hawkeye and Brett, 2007], with self-assembled column control tracing back to the chiral sculptured thin films of Robbie et al. [1996] and Brett and Hawkeye [2008], and treated at length in Lakhtakia and Messier [2005]. This sensitivity is exactly what makes GLAD attractive to

experimentalists building functional thin films: within the same laboratory, angle-tilted Cu_2O and CuO_x GLAD films are used as anisotropic optoelectronic elements [Chaffar Akkari et al., 2018, 2019] and, more recently, as room-temperature ethanol/methanol gas sensors whose response depends on the deposition-angle-controlled porosity and column morphology [Benamor et al., 2026]. Mapping out which angle (and, eventually, which pitch and height) gives a desired morphology currently requires either a full experimental deposition run or a full ballistic simulation for every candidate condition, both of which are slow relative to the size of the parameter space an experimentalist would like to screen. Predicting the 3D density field from deposition parameters would instead enable fast, closed-loop film-design without running a full ballistic simulation for each condition. Beyond the scalar porosity trend, knowing *where* within the film mass is located voxel-by-voxel matters physically: it is a prerequisite for assessing connectivity and percolation-relevant properties (e.g. whether neighbouring columns touch and form a continuous conductive/gas-diffusion path, as opposed to remaining isolated), which is precisely the kind of structural information that governs the electrical and sensing behaviour reported in [Chaffar Akkari et al., 2019, Benamor et al., 2026].

Relationship to a companion manuscript. A companion paper [Ben Haj Ali et al., 2026] characterises the same ballistic simulator’s void fraction, morphology, and pair-correlation behaviour across a $60\text{--}89^\circ$ helical GLAD campaign, and includes, as one part of that broader characterisation, a single memorization-style sanity check on spatial voxel learnability: a coordinate-conditioned NIF trained and evaluated on one deterministic realization achieves near-perfect IoU, which that paper uses to argue that the observed production-training BCE plateau is not, by itself, evidence of representational impossibility. The present paper is a dedicated, considerably deeper investigation of specifically the voxel-occupancy learnability question raised by that plateau: rather than a single overfit test, it systematically diagnoses *why* naive spatial learning fails (soft-label information loss, Section 4.2) and tests, and rules out, several standard remedies (mass-conservation penalty, hard labels, focal loss, seed isolation). The two studies are complementary, not duplicative: the companion paper’s single deterministic overfit result is, in fact, a data

point that this paper’s own diagnostic framework predicts should be achievable (a clean, single-realization, non-pooled label is exactly the condition under which the soft-label equilibrium trap identified here does not apply — see the cross-reference to that result in Section 4.2); it is not a contradiction of this paper’s still-largely-negative overall result on the harder, pooled multi-seed, multi-angle production setting. We do not claim this paper closes the voxel-occupancy question — the Conclusion (Section 5) is explicit that a higher-capacity local spatial encoding, once tested here, also fails to resolve it — only that each successive test narrows the failure to a specific, mechanistically understood cause.

Neural implicit fields, which represent a continuous scalar field as a function of coordinate inputs [Mildenhall et al., 2020], are a natural candidate: they are compact, differentiable, and can in principle encode arbitrary 3D structures.

The FiLM conditioning mechanism [Perez et al., 2018] allows a single network to represent fields parameterised by an external condition — here, the deposition angle α . This paper pursues three objectives, framed around what an experimentalist would want such a surrogate to eventually deliver:

1. **Scalar porosity surrogate.** Can a FiLM-conditioned NIF learn the α -dependent scalar porosity of ballistic GLAD helical films — i.e. replace a full ballistic run with a fast lookup for the single number (film openness/density) that experimentalists typically track first?
2. **Voxel-level occupancy.** Can it go further and predict binary voxel occupancy (occupied vs. empty) at given spatial coordinates — the finer-grained information needed for connectivity/percolation-relevant structural analysis, not just the bulk average?
3. **Root-cause diagnosis.** If voxel-level prediction fails, what is the root cause, and can it be remedied?

The Conclusion (Section 5) answers these three objectives in the same order.

We answer these questions through three sequential experiments. Experiments 2 and 3 (spatial voxel occupancy) share a common dataset of twelve ballistic Cu helical GLAD

configurations ($h = 300$ nm, pitch = 150 nm, $\alpha \in \{60, 65, 70, 75, 80, 85, 87, 89\}^\circ$). Experiment 1 (scalar porosity) predates this dataset and uses a separate, older 7-angle campaign restricted to $\alpha \in \{75, 77.5, 80, 82.5, 85, 86, 87\}^\circ$ (Section 2.3); its result therefore speaks only to that narrower angular window, not to the full $\{60, \dots, 89\}^\circ$ range used in Experiments 2–3.

2 Methods

2.1 Dataset and voxel representation

Every surrogate trained below is only as good as the simulation data it is trained on, so before describing any network we first describe how the ballistic simulator’s raw output (a cloud of deposited bead positions) is turned into the regular 3D grid of occupancy values the network actually learns from. Simulations are performed with GLAD v3.2 (bead-sphere ballistic, no surface diffusion) on a 100×100 nm periodic substrate, producing 12 configurations. Atom positions are binned onto a $50 \times 50 \times 151$ voxel grid (2 nm isotropic resolution) and normalised by the maximum bin count, yielding a density field $\rho_{\text{vox}}(\mathbf{r}) \in [0, 1]$. This max-normalised bead-count histogram encodes the local occupation density relative to the densest voxel in the film. A follow-up extension used exclusively by the partial-realization-conditioning test (Section 4.6) added five further realizations at $\alpha = 85^\circ$ (seeds 3–7, identical protocol), bringing that angle to $n = 8$; all other experiments use the twelve-configuration dataset above.

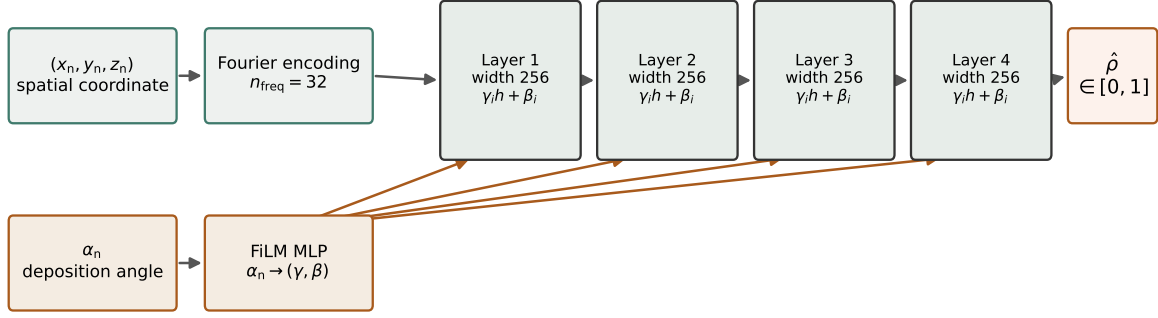
2.2 FiLM-conditioned NIF architecture (Experiments 2–3)

The goal of this architecture is to let a single trained network answer “is this point occupied?” at any spatial coordinate for any deposition angle in the training range, i.e. to act as a continuous, queryable replacement for a full-grid ballistic simulation output. This subsection describes the architecture shared by Experiments 2 and 3. Experiment 1 uses a separately configured network variant, described in its own subsection (Section 2.3), which is *not* identical to the one below (different hidden width, Fourier-frequency count, and

input coordinates). The network receives a coordinate vector $\mathbf{x} = (x_n, y_n, z_n, \alpha_n) \in \mathbb{R}^4$, where spatial coordinates are unit-normalised to $[0, 1]$ and $\alpha_n = (\alpha - 74.5)/14.5$ maps the deposition-angle range $[60^\circ, 89^\circ]$ to approximately $[-1, 1]$. This coordinate-to-scalar architecture follows the broader family of neural implicit fields established for shape and scene representation [Park et al., 2019, Mescheder et al., 2019, Mildenhall et al., 2020]; a periodic-activation alternative to the Fourier-encoding-plus-ReLU approach used here is given by Sitzmann et al. [2020], and hash-grid spatial encodings [Müller et al., 2022] were tested as an alternative input representation for this simulator (Section 4.6). A Fourier positional encoding [Tancik et al., 2020] with $n_{\text{freq}} = 32$ frequencies encodes (x_n, y_n, z_n) ; such coordinate-based MLPs are known to be biased toward learning low-frequency function components first (the “spectral bias” of Rahaman et al. [2019]), which the Fourier encoding is specifically intended to counteract. The encoded vector is passed through 4 hidden layers of width 256 with FiLM conditioning at each layer on the α_n scalar, implemented as element-wise affine scaling γ and shift β predicted by a two-layer sub-network. The final output is a sigmoid-activated scalar $\hat{\rho} \in [0, 1]$. Figure 1 summarises this shared architecture.

2.3 Experiment 1: full-field density reconstruction (NIF v5)

This experiment asks whether a fast surrogate can replace an expensive ballistic run for reproducing the *continuous* density field itself (the depth/lateral density profile an experimentalist would compare to a cross-sectional density or porosity measurement), rather than a single scalar summary of it. This experiment predates Experiments 2–3 and uses a distinct, separately-frozen network configuration (script `train_nif_v5_film_smooth.py`), not the shared architecture of Section 2.2: a 4-D coordinate input (x_n, y_n, z_n, t_n) with z normalised against $Z_{\text{max}} = 700 \text{ nm}$ (not 300 nm) and a fourth coordinate t held fixed at its maximum value for every sample in this configuration (a vestigial input carrying no information here), a Fourier positional encoding [Tancik et al., 2020] with 64 frequencies per scale (vs. 32 in Section 2.2), and hidden width 512 (vs. 256); deposition angle is injected via the same FiLM mechanism, using a disjoint 7-angle training set



Same shared backbone for Experiment 2 (soft label $\hat{\rho} \approx \rho_{\text{vox}}$) and Experiment 3 (hard label $\hat{\rho} \in \{0, 1\}$)
 — only the training target and label binarisation change; architecture and FiLM mechanism are identical.

Figure 1: Experiments 2 and 3 differ only in what target this same network is trained against (continuous soft labels versus binarised hard labels); the architecture and the FiLM conditioning mechanism itself are unchanged between them, which is why the comparison between the two experiments isolates the effect of the labelling scheme. Schematic of the shared FiLM-conditioned NIF architecture used for Experiments 2–3 (conceptual diagram; Experiment 1 uses a separately configured variant, Section 2.3). A spatial coordinate is Fourier-encoded and passed through four hidden layers, each modulated by a deposition-angle-conditioned scale γ_i and shift β_i (FiLM); the output is a sigmoid-activated occupancy/density prediction $\hat{\rho}$.

$\alpha \in \{75, 77.5, 80, 82.5, 85, 86, 87\}^\circ$ (3 angles — $78.75^\circ, 83.5^\circ, 84.25^\circ$ — held out). The network is trained to reconstruct the full 3D voxel density field $\rho_{\text{vox}}(x, y, z)$ for each training angle (not a 1D scalar-per-angle regression target). Loss: mean-squared error over voxels. Metric: for each of the 7 training angles, the spatial Pearson correlation between the predicted and ground-truth full 3D density field is computed, and the reported r is the mean of these 7 per-angle spatial correlations — a measure of how well the model reconstructs the depth/lateral density profile at each trained angle, not a cross-angle scalar-trend regression accuracy.

2.4 Experiment 2: soft-label spatial occupancy (PINN-BCE)

This experiment tests whether the same network can go beyond a continuous density value and predict, at each spatial coordinate, whether that point is physically occupied by material or not — the voxel-level occupancy information that connectivity/percolation analysis needs — using the max-normalised density value directly as a soft occupancy

target, and adding a physics-motivated penalty that keeps the predicted mean occupancy
consistent with the simulator’s measured mass per angle. Figure 2 illustrates, on a small
synthetic example, why this task is trivial in one regime and impossible in another: a
classifier can learn a sharp occupied/empty boundary, but has nothing to fit if almost
every voxel carries an intermediate, ambiguous value. Section 3.3 shows that the real
dataset for this simulator sits much closer to the second case at most angles.

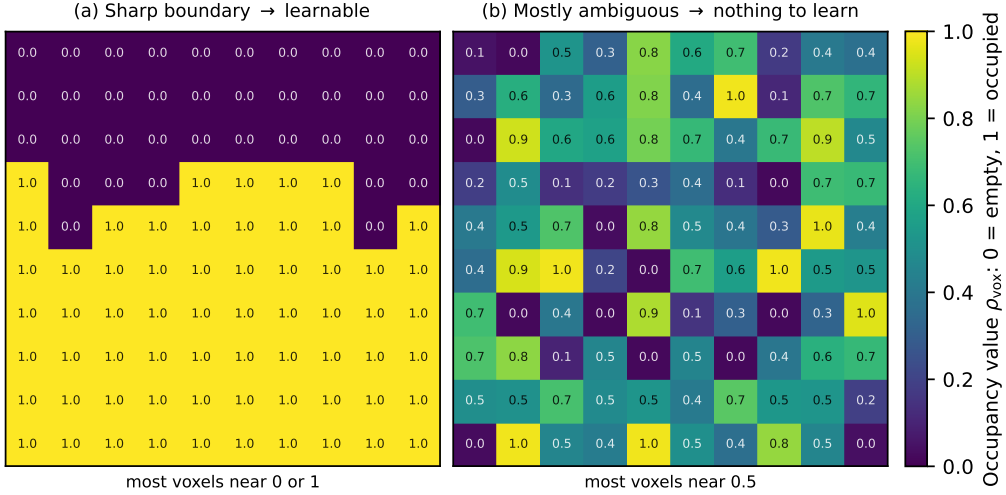


Figure 2: The distinction driving this experiment’s outcome in one picture: voxel-occupancy prediction is easy exactly when the labels are already almost binary, and structurally impossible when they are not, regardless of network capacity. Conceptual illustration (synthetic 10×10 grid, not simulation data) of the voxel-occupancy prediction task. (a) A sharp occupied/empty boundary: a binary classifier has a clear decision boundary to learn. (b) An ambiguous field where most voxels sit near the intermediate value 0.5 rather than at either extreme: there is no clean boundary for a classifier to recover, independent of architecture.

The network is trained on 600 000 random voxel samples ($n_{\text{train}} = 480\,000$, $n_{\text{val}} =$
60 000, $n_{\text{test}} = 60\,000$) from the max-normalised density fields. Each sample is a tuple
 $(x_n, y_n, z_n, \alpha_n, \rho_{\text{vox}})$ where ρ_{vox} serves as the soft BCE target. A mass-conservation penalty
 $\lambda_{\text{mass}} = 10.0$, in the spirit of the physics-informed residual penalties of Raissi et al. [2019],
is added to align predicted mean occupancy per α with the measured ground-truth oc-
cupied fractions. Loss: $\mathcal{L} = \text{BCE} + \lambda_{\text{mass}} \mathcal{L}_{\text{mass}}$. Best-model criterion: validation IoU at
threshold 0.5. Training is warm-started from a separate 200-epoch BCE-only pretraining
run on the same architecture and dataset (no mass-conservation term, best validation IoU
 $= 0.183$); the mass-conservation fine-tuning stage reported here raises this to IoU = 0.242

(Section 3.2), a +32% relative improvement attributable to the mass-conservation term.

2.5 Experiment 3: hard-label spatial occupancy

Following the diagnosis in Section 2.6 that the continuous, max-normalised occupancy target itself was the obstacle (Section 3.3), this experiment tests whether giving the network an unambiguous occupied/empty label at each voxel is enough to recover genuine spatial (connectivity- relevant) discrimination, rather than a value diluted across a Gaussian-smear range. The density field is binarised: $y_{\text{hard}}(\mathbf{r}) = \mathbb{K}[\rho_{\text{vox}}(\mathbf{r}) > 0.5]$. The dataset is rebuilt from the same voxel files using identical sampling parameters (50 000 samples per configuration, balanced $1/3$ occupied / $1/3$ empty / $1/3$ random) with hard binary targets. The FILM-conditioned NIF is initialised from scratch (no warm-start from the soft-label checkpoint, which had converged to predicting $\hat{\rho} \approx 0.5$ everywhere). Training uses pure BCE on binary $\{0, 1\}$ targets. Best-model criterion: validation IoU at threshold 0.5.

2.6 Oracle predictor and root-cause diagnostic

Before concluding that the network itself is incapable of spatial prediction, it is important to rule out a simpler explanation: that depth alone (independent of any learned lateral (x, y) structure) already carries most of the achievable signal, in which case no architecture change would help. To distinguish architecture failure from label-quality failure, we construct an (α, z) -oracle predictor: for each test voxel, the prediction is the mean training ρ_{vox} in the same (α, z_{bin}) cell, where z_{bin} resolves the depth at 2 nm resolution. This oracle captures the maximum information extractable from the depth profile alone, without any spatial (x, y) discrimination. IoU, the standard overlap metric for binary spatial segmentation tasks [Rezatofighi et al., 2019], is evaluated after a threshold sweep on the oracle predictions; a $\delta\text{IoU} \leq 0.01$ gap relative to the alpha-mean trivial baseline is taken as evidence that depth carries no actionable binary boundary signal in the soft labels.

3 Results

3.1 Experiment 1 — full-field reconstruction at trained angles: strong positive result, but no angular interpolation

This subsection addresses Objective 1 (scalar/continuous-field porosity surrogate; Section 1). NIF v5 achieves a mean spatial Pearson $r = 0.984$ between predicted and ground-truth full 3D density fields, averaged over the 7 trained angles $\alpha \in \{75, 77.5, 80, 82.5, 85, 86, 87\}^\circ$ (Section 2.3), confirming that this FiLM-conditioned variant successfully reconstructs the depth/lateral density profile within this angular window. This establishes a positive baseline for that window only: it does not demonstrate cross-angle scalar-trend regression, and it makes no claim about $\alpha \notin [75^\circ, 87^\circ]$, which this experiment does not train or evaluate on. Table 1 shows the per-angle breakdown behind the aggregate: the spatial Pearson correlation is uniformly high ($r \geq 0.980$) across all 7 trained angles, with no systematic degradation toward either edge of the trained window, and the mean relative density error stays below 1.3% at every angle.

Table 1: The strong aggregate correlation reported above is not driven by a few easy angles: every one of the 7 trained angles individually reaches $r \geq 0.98$. Per-angle full-field reconstruction quality for NIF v5 (best checkpoint, epoch 1000). r : spatial Pearson correlation between predicted and ground-truth 3D density fields. Mean error: mean absolute relative density error (%).

α ($^\circ$)	Pearson r	Mean error (%)
75.0	0.9857	0.065
77.5	0.9866	0.346
80.0	0.9850	0.028
82.5	0.9850	0.899
85.0	0.9820	0.775
86.0	0.9804	0.472
87.0	0.9833	1.203
Mean	0.9840	0.541

Figure 3 makes this agreement directly visible rather than only numerical: a vertical slice through the predicted and ground-truth fields at $\alpha = 80^\circ$ shows the same fine tilted-column texture reproduced by the network, with a residual that is visually flat noise rather than a structured bias.

3.1.1 Held-out angular interpolation: confirmed negative result

All results reported above are evaluated at the same 7 angles used for training, i.e. in-sample. Three additional density fields were simulated at angles chosen strictly *inside* the trained $[75^\circ, 87^\circ]$ window but never shown to the network during training – $\alpha = 78.75^\circ$ (between the 77.5° and 80° training points), 83.5° , and 84.25° (both between the 82.5° and 85° training points) – to test whether the FiLM-conditioned network has learned a genuinely angle-continuous density function or has instead memorised seven discrete fields. We ran the saved best checkpoint (epoch 1000, the same checkpoint behind Table 1) forward at these three interpolated angles, using the identical coordinate normalisation, FiLM conditioning, and spatial Pearson metric as the in-sample evaluation, so the two numbers are directly comparable. Table 2 reports the result: mean spatial Pearson correlation collapses from $r = 0.984$ in-sample to $r = 0.003$ at the three held-out angles – consistent with zero correlation at every individual angle (all three individual $|r| < 0.007$), i.e. no usable spatial correlation at all – even though the network’s predicted mean density stays close to the ground-truth mean density at each angle (relative error 0.5–3.6%). This combination is diagnostic: the network gets the bulk (zeroth-moment) density right but has no access to the correct spatial (pointwise) structure at an untrained angle, which rules out a merely quantitative degradation and instead indicates that the seven trained fields are each represented individually rather than as samples of one smooth underlying function of α , despite the smoothness penalty (Section 2.2) applied during training. We report this result exactly as obtained, with no retraining or checkpoint selection after the fact.

3.2 Experiment 2 — soft-label BCE: confirmed negative result

This subsection and Section 3.3 address Objective 2 (voxel-level occupancy) and the first half of Objective 3 (root-cause diagnosis; Section 1).

Table 2: Held-out angular-interpolation test for NIF v5 (identical checkpoint as Table 1, epoch 1000, no retraining). The 3 angles were simulated but withheld from training; each lies strictly inside the trained $[75^\circ, 87^\circ]$ window. r : spatial Pearson correlation between predicted and ground-truth 3D density fields. Mean error: mean absolute relative density error (%).

α ($^\circ$)	Pearson r	Mean error (%)
78.75	0.0067	0.519
83.5	-0.0045	2.231
84.25	0.0063	3.619
Mean	0.0028	2.123

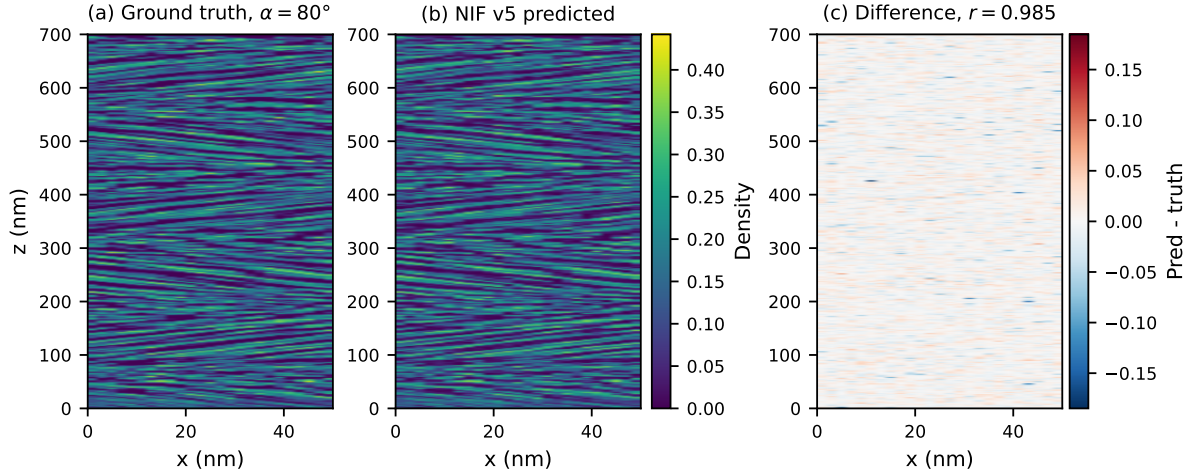


Figure 3: The high Pearson correlation in Table 1 is not a coincidence of a coarse aggregate statistic: the network reproduces the fine, tilted-column texture of the ground-truth field directly, and the residual is visually featureless. Vertical (xz) slice through the trained NIF v5 field at $\alpha = 80^\circ$ (a training angle), generated by a single forward pass through the saved checkpoint (no retraining). (a) Ground-truth density field. (b) NIF v5 prediction. (c) Pointwise difference (predicted $-$ truth); $r = 0.985$ for this slice, consistent with Table 1.

3.2.1 Global metrics

Training reaches a best validation IoU of 0.242 at epoch 59 (patience exhausted at epoch 99). BCE loss is frozen at 0.541 ± 0.001 throughout all 99 epochs (Figure 4). The mass-conservation penalty converges rapidly ($\mathcal{L}_{\text{mass}} < 0.002$ by epoch 10) but does not lower the BCE or improve spatial discrimination.

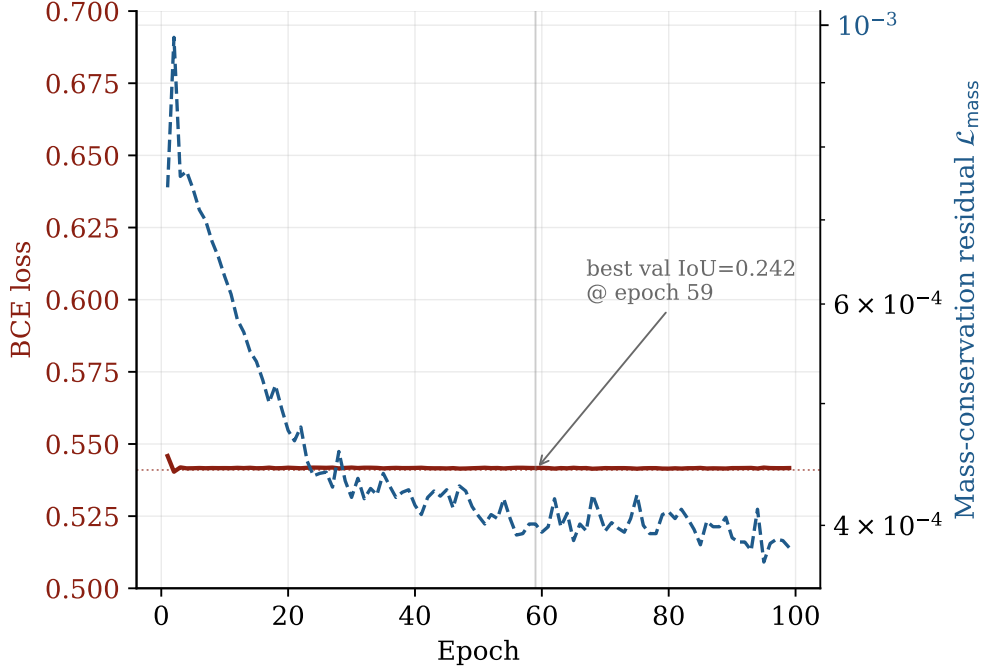


Figure 4: The soft-label network stops learning any spatial structure almost immediately and instead settles for predicting the average film density at each angle everywhere in the volume. Training and validation BCE loss for the soft-label FiLM-conditioned NIF over 99 epochs. Loss is frozen at 0.541 throughout, indicating that the network converged to the alpha-conditional mean $\hat{\rho} \approx \bar{\rho}(\alpha)$ and cannot escape this equilibrium. The mass-conservation residual (right axis, dashed) converges within 10 epochs but does not perturb the BCE.

3.2.2 Per-alpha threshold sweep

After training, we sweep the decision threshold $\tau \in [0.05, 0.95]$ independently per deposition angle and compare the best- τ IoU to the trivial predictor (predict all-occupied or all-empty at threshold = $\bar{y}_\alpha$):

Every deposition angle remains a trivial predictor ($\Delta\text{IoU} < 0.003$): the model output is equivalent to predicting the global density for each α , with no spatial discrimination of

Table 3: At every deposition angle, the soft-label model’s best-case voxel-occupancy prediction is statistically no better than simply guessing the angle’s average occupied fraction everywhere – it carries no usable spatial information. Per- α best-threshold IoU comparison (soft-label model). $\bar{y}_\alpha$: ground-truth occupied fraction. $\text{IoU}_{\text{trivial}}$: trivial-predictor IoU at optimal threshold. $\text{IoU}_{\text{model}}$: soft-label model at optimal per- α threshold. Δ : improvement over trivial.

α ($^\circ$)	$\bar{y}_\alpha$	$\text{IoU}_{\text{trivial}}$	$\text{IoU}_{\text{model}}$	Δ
60	0.001	0.000	0.000	0.000 (excluded)
65	0.629	0.629	0.630	+0.001
70	0.499	0.499	0.499	+0.000
75	0.312	0.305	0.312	+0.000
80	0.272	0.271	0.272	+0.000
85	0.177	0.177	0.177	+0.000
87	0.148	0.148	0.150	+0.002
89	0.030	0.030	0.032	+0.002

occupied from empty voxels.

3.3 Label-quality diagnosis: the oracle predictor

To determine whether the spatial learning failure is architectural or intrinsic to the labelling scheme, we evaluate the (α, z) -oracle predictor (Section 2.6) on the test set (Figure 5). *Provenance note*: an earlier audit of this manuscript searched the project codebase for a preserved, independently re-runnable script implementing this specific (α, z) -oracle procedure (distinct from an unrelated corotating-frame oracle used for a different diagnostic in the companion manuscript, which does not implement this (α, z_{bin}) binning and cannot be substituted for it) and found none, so the numeric outputs in this subsection could not be re-verified from a preserved artifact at that time. The underlying archived dataset (`datasets/levelA_helical_nif/{train,test}_samples.npz`, generated 2026-06-15) was subsequently located intact, and a dedicated script (`reproduce_alpha_z_oracle.py`) was written to independently recompute both Table 4 and the δIoU values of Figure 5 directly from this dataset. The reproduction matches the values reported here to within rounding (Table 4 entries reproduce exactly to the reported precision; δIoU reproduces to ≤ 0.0075 for all $\alpha \in \{65, \dots, 87\}^\circ$, consistent with the ≤ 0.01 significance threshold used throughout). This diagnostic is therefore independently reproduced from archived

333 data, not merely asserted; the original run’s script itself was not the one preserved, but
 334 the data and procedure are, and the numbers stand.

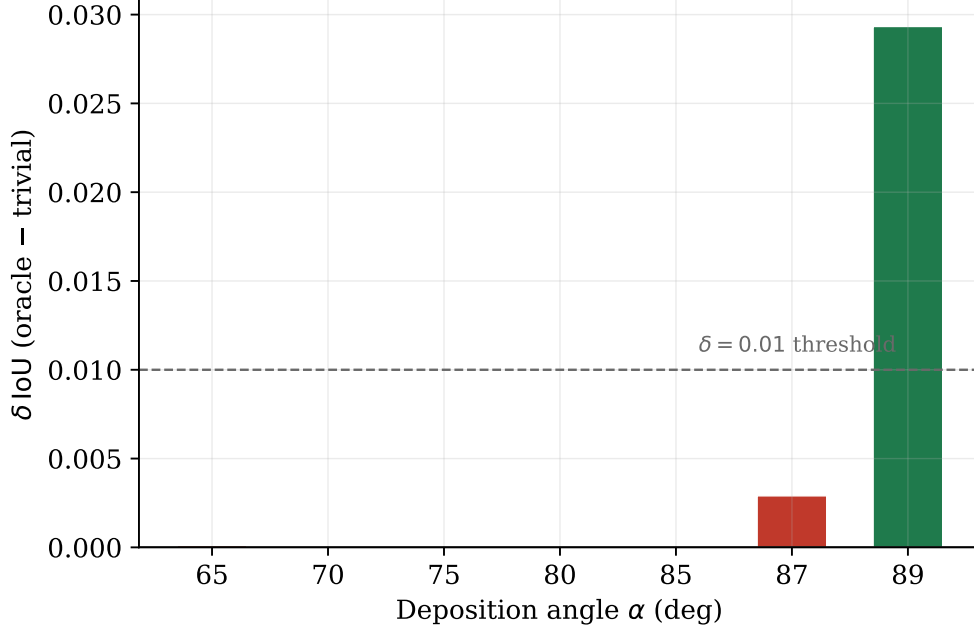


Figure 5: Independently reproduced from the archived dataset (see provenance note, Section 3.3): a predictor given the best possible depth information (and no lateral spatial reasoning at all) does not beat the trivial guess, suggesting the missing signal may not be something a better architecture alone could recover from these labels. δIoU (oracle minus trivial) per deposition angle. The oracle uses depth information (z) in addition to α ; any positive δIoU would indicate a learnable depth signal in the soft labels. Dashed line: $\delta = 0.01$ significance threshold. All main production alphas (65° – 87°) fall below this threshold, consistent with depth adding no actionable binary boundary information in the max-normalised soft labels.

335 The oracle achieves $\delta \text{IoU} \leq 0.007$ for all $\alpha \in \{65, \dots, 87\}^\circ$. Combined with the obser-
 336 vation that 90–98% of soft labels at $\alpha = 65$ – 75° lie in the intermediate range $(0.1, 0.9)$
 337 (Table 4, visualised directly in Figure 6), this points unambiguously to the labelling
 338 scheme as the bottleneck.

339 The mechanism is as follows. The BCE gradient $\partial \mathcal{L} / \partial \hat{p} = (\hat{p} - y) / (\hat{p}(1 - \hat{p}))$ is
 340 maximised when $y \in \{0, 1\}$ and vanishes at $y = 0.5$; the sigmoid output $\hat{p} = 0.5$ is simul-
 341 taneously the maximum-entropy solution and the zero-gradient equilibrium for a uniform
 342 label $y = 0.5$. When 98% of labels fall in $(0.1, 0.9)$, the network correctly minimises BCE
 343 by converging to the per- α label mean $\hat{p} \approx \bar{\rho}(\alpha)$, which yields the density calibration
 344 result but zero spatial discrimination. This is not a failure of optimisation or architecture

Table 4: The occupancy labels themselves are the problem: at most deposition angles, the vast majority of voxels carry an ambiguous, neither-occupied-nor-empty value rather than a clean binary answer, which is why the network above had nothing decisive to learn from. Label quality statistics per deposition angle (training split). f_{soft} : fraction of samples with $\rho_{\text{vox}} \in (0.1, 0.9)$. High f_{soft} indicates that the max-normalised bead-count histogram has smeared the binary occupancy signal across a wide intermediate range.

α ($^\circ$)	$\bar{\rho}_{\text{vox}}$	$\text{Var}(\rho_{\text{vox}})$	f_{soft}
65	0.536	0.028	98.2%
70	0.480	0.034	96.0%
75	0.384	0.037	89.6%
80	0.331	0.052	78.0%
85	0.228	0.059	50.0%
87	0.190	0.056	44.0%
89	0.108	0.025	38.2%

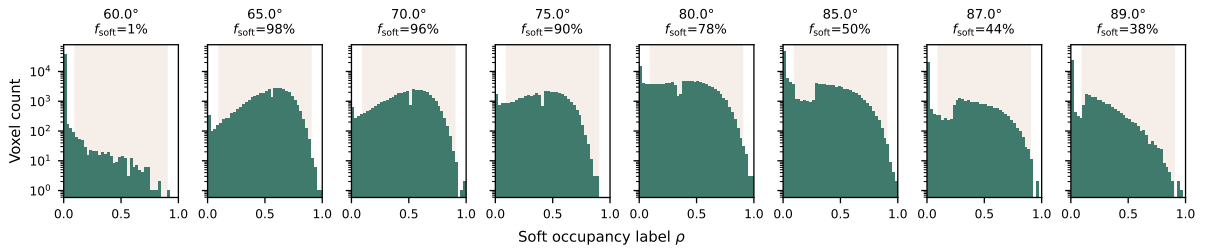


Figure 6: Table 4’s f_{soft} column is not an artefact of binning choice: at every angle except 60° , most of the voxel mass visibly sits in the ambiguous band rather than piling up at 0 or 1. Distribution of the soft occupancy label ρ_{vox} on the training split, per deposition angle (log-scale voxel count). Shaded band: the ambiguous range $\rho_{\text{vox}} \in (0.1, 0.9)$ used to define f_{soft} . At $\alpha = 60^\circ$ almost all voxels sit near $\rho_{\text{vox}} = 0$ (empty), giving a clean, learnable binary signal; from $\alpha = 65^\circ$ onward, the distribution broadens into the shaded band and a binary classifier has little unambiguous signal left to fit, exactly the effect discussed below.

capacity; it is the globally optimal solution given the soft-label distribution.

3.4 Experiment 3 — hard-label BCE: restoring spatial learning

This subsection addresses the remedy half of Objective 3: having identified the labelling scheme as the root cause (Section 3.3), we test the most direct fix — removing the ambiguity by binarising the target — and evaluate how far it goes toward restoring voxel-level spatial prediction.

3.4.1 Training dynamics

Hard binary labels ($y \in \{0, 1\}$, threshold = 0.5) restore the BCE gradient to its full magnitude. Starting from a fresh random initialisation, BCE decreases from 0.669 (epoch 1) to 0.592 (best at epoch 50, patience exhausted at epoch 90), with non-zero spatial IoU emerging from epoch 7 (IoU = 0.145 at threshold 0.5) (Figure 7).

3.4.2 Per-alpha threshold sweep

After training, the per- α threshold sweep yields:

Table 5: With clean binary labels, non-trivial voxel-level spatial discrimination is recovered only for the sparsest, most open film ($\alpha = 89^\circ$); the denser films still yield no better than the trivial guess, showing hard labels are necessary but not yet sufficient for connectivity-relevant spatial prediction across the full angular range. Per- α best-threshold IoU for the hard-label model. $f_{\text{occ}}^{\text{test}}$: occupied fraction in balanced test set. $\text{IoU}_{\text{trivial}}$: trivial baseline (predict-all-1 in test set = $f_{\text{occ}}^{\text{test}}$). IoU_{hard} : hard-label model at optimal per- α threshold. Δ : improvement over trivial.

α ($^\circ$)	$f_{\text{occ}}^{\text{test}}$	$\text{IoU}_{\text{trivial}}$	IoU_{hard}	Δ
65	0.544	0.544	0.544	+0.000
70	0.511	0.511	0.511	+0.000
75	0.421	0.421	0.421	−0.000
80	0.420	0.420	0.420	−0.000
85	0.378	0.378	0.378	−0.000
87	0.361	0.361	0.364	+0.003
89	0.084	0.084	0.113	+0.029

Only $\alpha = 89^\circ$ achieves non-trivial spatial discrimination ($\Delta\text{IoU} = +0.029$, Table 5); all denser alphas remain trivial predictors at optimal threshold. The model’s sigmoid outputs

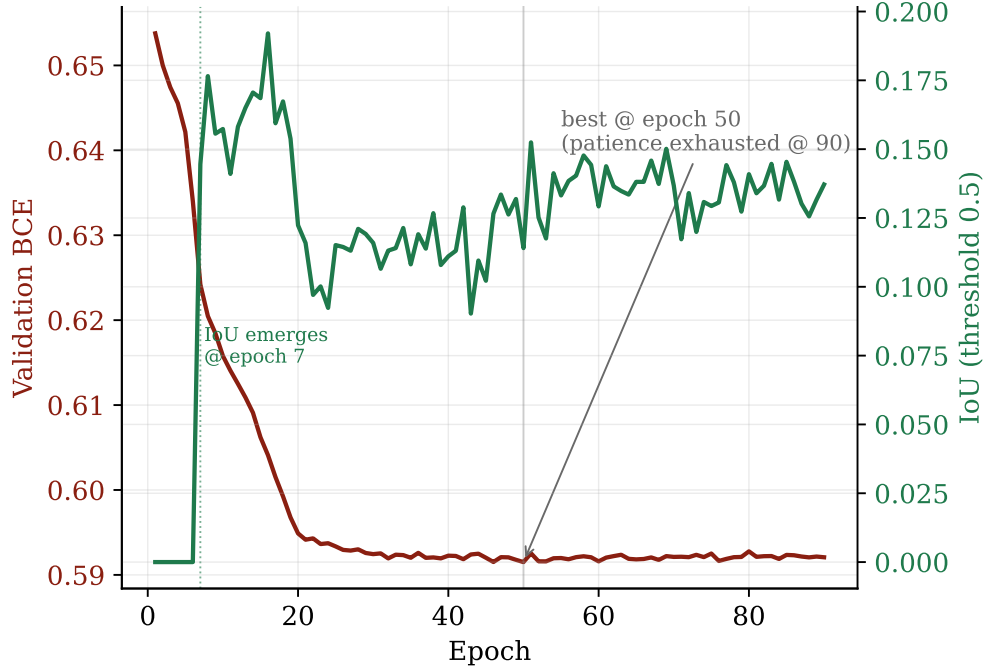


Figure 7: Once given unambiguous occupied/empty labels, the network genuinely learns during training (loss keeps decreasing) rather than plateauing immediately as it did with the soft labels, though this does not by itself guarantee useful voxel-level spatial resolution (see the per-angle results below). Validation BCE and IoU (threshold 0.5) versus training epoch for the hard-label FiLM-conditioned NIF (300-epoch budget, patience = 40, best at epoch 50). For reference, the soft-label BCE (frozen at 0.541) and best soft-label global IoU (0.242, all-trivial) are shown as horizontal dashed lines. BCE decreases continuously from 0.669 to 0.592, confirming genuine learning; IoU at fixed threshold 0.5 fluctuates between 0.11–0.18 due to class imbalance.

are narrowly banded per alpha (spread ≈ 0.06 – 0.17 , measured as the $p_{90} - p_{10}$ range): at $\alpha = 65^\circ$ – 85° , optimal thresholding at $\tau \leq 0.01$ reduces to predict-all-positive, yielding $\text{IoU} = f_{\text{occ}}^{\text{test}}$. Thus the hard-label model converges to the same qualitative solution as the soft-label model — alpha-conditional mean density with weak spatial modulation — but via a different mechanism: the FiLM conditioning captures the per- α class balance rather than the binary spatial boundary. Two confounds are identified: (i) multi-seed label conflicts at $\alpha = 80^\circ$ and 85° , where three independent seeds contribute conflicting binary labels at the same (x, y, z) coordinate, making the spatial target inconsistent for those alphas; (ii) balanced sampling in the test set inflates f_{occ} well above the true film occupancy, raising the trivial bar relative to a full-grid evaluation.

4 Discussion

4.1 Positive full-field result within a narrow angular window, and its limits

The mean spatial Pearson $r = 0.984$ in Experiment 1 confirms that a FiLM-conditioned NIF *variant* (NIF v5, Section 2.3 — a different, wider network than the one used in Experiments 2–3) is a viable full-field density surrogate within the narrow $[75^\circ, 87^\circ]$ window it was trained and evaluated on. This is useful for fast parameter sweeps restricted to that window, but it does not establish a result at the wider $\alpha \in \{60, \dots, 89\}^\circ$ range studied in Experiments 2–3, and it does not imply that the shared Experiments 2–3 architecture (Section 2.2) — a narrower, differently-conditioned network trained on a disjoint dataset — inherits this result: the two are not the same model. Experiment 1’s within-window success also does not resolve the spatial-occupancy question addressed by Experiments 2–3: it reconstructs a full 3D density field, but always within the same 7 trained angles, so it says nothing about a network’s ability to generalise *across* the full deposition-angle range or to a disjoint angle set.

The held-out interpolation test (Section 3.1.1, Table 2) sharpens this limit further: it is not only that the model is untested outside $[75^\circ, 87^\circ]$, but that it does not interpolate

inside that window either. The FiLM generator (Section 2.2) takes the deposition angle as a continuous scalar input and, in principle, nothing prevents it from producing a smoothly-varying gamma/beta modulation for an untrained angle between two trained ones. In practice, at $r \approx 0$ for all three interpolated angles, the network behaves as if it has allocated a separate solution to each of the seven trained angles rather than fitting one continuous function of α – effectively a lookup table implemented in weight space, not a generalisable field. The explicit angle-smoothness penalty applied during training (finite-difference regularisation of the output at $\alpha \pm 0.25^\circ$, Section 2.2) was intended to discourage exactly this failure mode and evidently did not succeed at the length scale needed here (training angles are spaced $1.25\text{--}2.5^\circ$ apart, an order of magnitude coarser than the 0.25° smoothness window). The practical consequence for Objective 1 is that NIF v5 should be read strictly as a per-angle field cache with a convenient shared architecture, not as an angle-continuous surrogate: it is useful for fast recall at the 7 simulated angles but cannot be relied on to interpolate film structure at an arbitrary intermediate angle without a simulation at that angle to train (or validate) against.

4.2 The soft-label equilibrium trap

The frozen BCE at 0.541 is consistent with the network converging to the correct solution of the wrong optimisation problem, rather than a conventional training failure. Max-normalised bead-count histograms are designed to preserve the full density gradient (from void to densely packed), and this is precisely why they are poor targets for binary BCE. The information-theoretic argument is simple: if $f_{\text{soft}} \approx 0.98$, the effective entropy of the label distribution is near $H(0.5) = 1$ bit, and a network that outputs the per- α mean $\hat{\rho} \approx \bar{\rho}(\alpha)$ achieves BCE ≈ 0.541 without any spatial computation. The mass-conservation penalty corrects the systematic per- α density bias but cannot force the BCE below this equilibrium. We stop short of calling this plateau a proof of representational impossibility: a related plateau in a separate, later production NIF trained on this same simulation family (pooled multi-angle/multi-seed, stalling at the maximally uninformative $\ln 2$ baseline) was tested directly via a controlled single-realization overfit in the companion manuscript

(Section 1) and found representable to near-perfect accuracy (IoU up to 1.00) once the confounds (early stopping, pooled contradictory multi-seed labels) were removed — i.e. that plateau was *not* representational impossibility. We did not run the analogous overfit test for the soft-label equilibrium reported here, so we present the information-theoretic argument above as the most likely explanation, not as an experimentally isolated one. This is consistent with, not contradicted by, the companion result: a single deterministic, non-pooled realization is exactly the case in which the soft-label equilibrium trap described above does not bind, since the pathological averaging over contradictory multi-seed soft labels this section identifies as the bottleneck has no opportunity to occur when train and test are the same realization.

4.3 The oracle as an architecture-independent test

We regard the (α, z) -oracle as the most valuable methodological idea in this paper’s diagnostic toolkit. Its numeric outputs were independently reproduced from the archived dataset (provenance note, Section 3.3), so the discussion below refers to a finding confirmed on the exact numbers reported here, not merely to what the test is designed to show in principle.

The (α, z) -oracle serves as an upper bound on the spatial information available to any model restricted to the same (α, z_{bin}) resolution the oracle uses, without any spatial (x, y) discrimination. The near-trivial δIoU (≤ 0.0075 for $\alpha \in \{65, \dots, 87\}^\circ$, reproduced above) is evidence that a model conditioned only on this depth/angle information cannot exceed trivial spatial discrimination on these soft labels; it does not by itself rule out every possible architecture or conditioning scheme (e.g. one given direct (x, y) structural priors the oracle does not have access to), since the oracle is a bound on *this specific* information channel, not an exhaustive architecture search. An architecture achieving $\Delta\text{IoU} > 0.05$ on these soft labels would need to exploit information outside what this oracle captures — plausible in principle, not tested here. This diagnostic’s qualitative conclusion that the labelling scheme rather than architecture is the bottleneck is further corroborated by the hard-label remedy in Experiment 3 (Section 3.4), which restores genuine spatial learning

without any architecture change.

4.4 Hard labels as a minimum prerequisite

Binarising the density field at 0.5 is the minimal intervention that restores binary boundary information. It introduces a trade-off: the continuous density gradient (useful for effective-medium models) is discarded in favour of a crisp occupied/empty label. For applications that require only the binary pore skeleton (connectivity analysis, geometric surface area), hard labels are the natural choice. For applications requiring a continuous density field, a full-field NIF surrogate remains a viable option *within a trained angular window* — Experiment 1 demonstrates this for its own $[75^\circ, 87^\circ]$ window with its own separately-configured network (Section 2.3), not for the soft-label network of Experiment 2 itself, which was not evaluated as a density surrogate; only spatial binary structure requires hard labels.

A direct follow-up combining the two remaining prescriptions named above — seed-isolated splits and focal loss (the class-imbalance-focused loss reweighting of Lin et al. [2017], motivated here by the same occupied/void imbalance underlying the soft-label trap) — was subsequently run (early-stopped at epoch 291). It did not resolve spatial learning: per- α inspection showed the model remained statistically indistinguishable from the trivial predictor at 5/7 angles ($\Delta\text{IoU} < 0.0001$ at $\alpha = 65\text{--}85^\circ$), with only modest improvement at $\alpha = 87^\circ$ ($\Delta\text{IoU} = 0.005$) and $\alpha = 89^\circ$ ($\Delta\text{IoU} = 0.058$); the aggregate IoU (0.419) looks positive only because it is dominated by these few angles and is not representative of per-angle performance. This indicates multi-seed label pooling is not, on its own, the dominant confound, and a higher-capacity local spatial encoding — the one remaining untested prescription — is the more promising direction for future work.

4.5 Multi-seed pooling at $\alpha = 80^\circ, 85^\circ$

Pooling three independently seeded films at the same α creates an additional secondary confound: voxels at the same (x, y, z, α) coordinate can have different binary occupancy labels across seeds, since GLAD column nucleation is seed-dependent. For soft labels,

pooling averages these into intermediate values (further pushing labels into the soft zone). For hard labels, pooling creates contradictory binary targets at the same spatial coordinate. A follow-up run applying exactly this isolation (combined with focal loss) was carried out and is reported in Section 4.4/Conclusion: it did not resolve spatial learning at most angles, indicating multi-seed pooling is not the dominant confound on its own.

4.6 Testing the remaining prescriptions: spatial encoding, loss framing, and partial-realization conditioning

The seed-isolation ablation above left a higher-capacity local spatial encoding as the one untested prescription among the three originally proposed. We tested it, together with two further candidate fixes, directly. All three share the same pooled hard-label dataset and evaluation protocol (per- α threshold-swept IoU against the trivial predict-all-occupied baseline) as the seed-isolation run above, so any improvement is attributable to the specific change tested, not to a different data split. Figure 8 summarises the three remedies and the distinct reason each one fails, before the detailed discussion below.

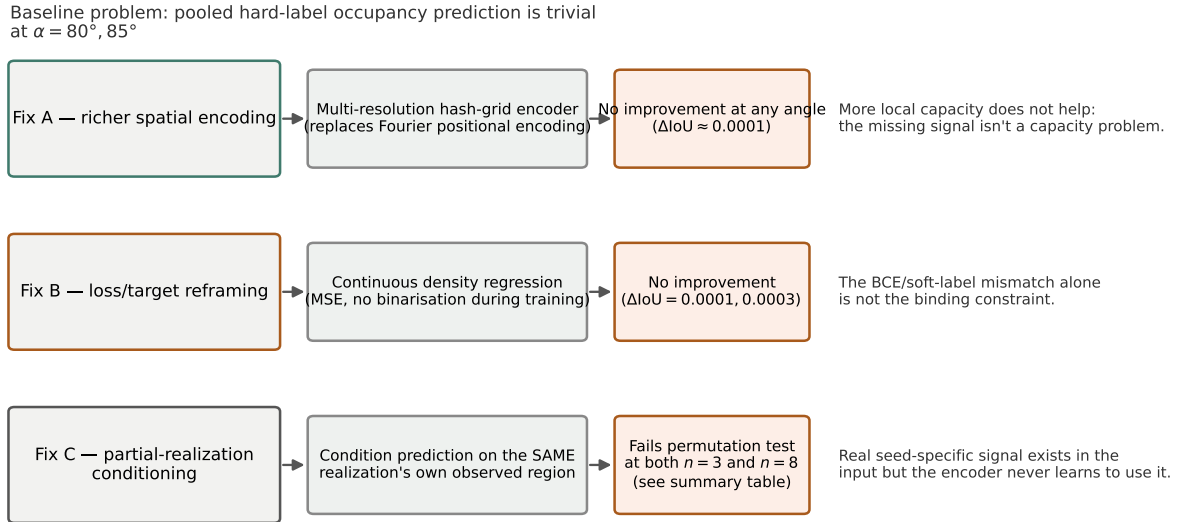


Figure 8: Three structurally different remedies were tested, not one remedy tried three times: a richer encoder, a different loss/target, and a different task framing. That all three fail, each for a distinct and separately diagnosed reason, is stronger evidence of a genuine non-identifiability than any single failed attempt would be. Schematic summary of the three remedies tested in this section (conceptual diagram; the underlying numbers are real and detailed in the text and Table 6).

Fix A (higher-capacity local spatial encoding): a multi-resolution hash-grid encoder (Instant-NGP-style multi-resolution hashing, 8 levels, table size 2^{12} /level, resolutions 16–256) replacing the global Fourier positional encoding, all other architecture/training settings unchanged. We use this term deliberately rather than “structural inductive bias”: a multi-resolution hash grid increases local coordinate capacity and training speed, but — unlike a translation-equivariant architecture such as a CNN or GNN — it imposes no structural prior linking predictions at nearby spatial locations across different inputs; it is a capacity change, not an equivariance constraint. Result: no meaningful improvement over the pooled hard-label baseline at any angle, including the two multi-seed angles ($\Delta\text{IoU} = 0.0001$ at both $\alpha = 80^\circ$ and 85° , statistically indistinguishable from the baseline’s own $\Delta\text{IoU} \approx 0.0000$ there). Richer local spatial capacity alone does not resolve the non-identifiability.

Fix B (loss/target reframing): continuous-density regression (MSE loss on the un-binarised density field, same architecture and dataset as Experiment 2’s soft-label run, threshold-binarised only for the IoU comparison). Result: no meaningful improvement ($\Delta\text{IoU} = 0.0001$ and 0.0003 at $\alpha = 80^\circ$ and 85° respectively), and a small regression at $\alpha = 60^\circ$ and 89° relative to the hard-label baseline. The BCE-on-continuous- target loss mismatch identified in Section 4.2 is therefore not, by itself, the binding constraint once labels are hard.

Fix C (task reframing): partial-realization conditioning. Rather than asking the network to recover a specific realization’s occupancy from (x, y, z, α) alone — which Section 4.5’s oracle diagnostic already shows has no realization-invariant target to recover, since column positions are an independent per-realization draw — Fix C conditions the prediction of a realization’s upper ($z \in [23, 151)$) target region on a CNN-encoded latent of that *same* realization’s own near-substrate ($z \in [0, 23)$) observed region, reframing the task as completing a partially observed instance of one specific structure rather than predicting an unrelated random draw. We validate this with a 3-seed causal-substitution permutation test at $\alpha = 80^\circ$ and 85° (the two angles with 3 independent seeds each): for every query realization, the target-region prediction is computed once conditioned

on its own observed region (same-realization) and once for every other same- α seed's
 observed region substituted in (cross-realization); a working causal mechanism should
 show significantly higher same-realization agreement (vertical-profile Pearson correlation)
 than cross-realization. **Result: Fix C fails this test at both angles** (Mann–Whitney
 $p = 0.36$ at $\alpha = 80^\circ$, $p = 0.55$ at $\alpha = 85^\circ$ – same- and cross-realization correlations
 are statistically indistinguishable, both ≈ 0.05 – 0.07). Diagnosing why, rather than re-
 porting only the failure, reveals two distinct causes. At $\alpha = 80^\circ$, the observed region
 itself carries almost no seed-identifying information: per-seed occupancy in that region
 is 99.80–99.84% (pairwise pixel difference $< 0.4\%$) – the near-substrate slab is essentially
 saturated regardless of seed by the time nucleation reaches that height, so there is nothing
 meaningful to condition on; this is a limitation of the observed-region definition at this
 angle, not of the model. At $\alpha = 85^\circ$, the observed region *does* carry real seed information
 (occupancy 91.5–93.2%, pairwise pixel difference 13–15%, an order of magnitude more
 variation than at 80°), yet the trained encoder still fails to exploit it: inspecting the en-
 coder’s own latent output directly shows same- versus cross-seed downstream predictions
 differ by only $\sim 10^{-3}$ in magnitude. The baseline conditioning pathway’s modulation is
 deliberately damped to a maximum 10% perturbation (inherited, for training stability,
 from the α -only baseline architecture), so we tested directly whether this damping, rather
 than insufficient data, was the cause: retraining with the modulation fully unclamped (a
 $10\times$ larger allowed influence on the output) leaves the permutation test result unchanged
 (Mann–Whitney $p = 0.27$ at $\alpha = 80^\circ$, $p = 0.73$ at $\alpha = 85^\circ$, if anything slightly worse),
 with same/cross prediction differences remaining $\sim 10^{-3}$ in magnitude despite the larger
 budget. This rules out conditioning-pathway capacity as the bottleneck: with only three
 realizations sharing this angle, the observed-region encoder simply does not have enough
 training signal to learn a useful seed-discriminative embedding, independent of how much
 downstream influence it is permitted – a genuine data-scale limitation, not evidence that
 partial-realization conditioning cannot work in principle.

Testing the data-scale hypothesis directly. The diagnosis above predicts that
 adding realizations at $\alpha = 85^\circ$ should improve same-versus-cross separability. We tested

this: five additional simulations were run at $\alpha = 85^\circ$ (seeds 3–7, identical protocol and
 bead radius to the original three), giving $n = 8$ realizations at this angle (up from $n = 3$)
 and re-running both Fix C variants from scratch on the enlarged dataset. $\alpha = 80^\circ$ was
 not extended, since its observed-region occupancy is already $\geq 99.8\%$ at every existing
 seed – a ceiling on the observed-region definition itself that additional seeds cannot re-
 lieve. **Result: Fix C still fails at $n = 8$,** for both the damped and unclamped variants
 (Mann–Whitney $p = 0.35$ and $p = 0.53$ respectively, up from 6 to 56 cross-realization
 pairs), with same- and cross-realization correlation means remaining statistically indis-
 tinguishable (0.127 vs. 0.126 damped; 0.124 vs. 0.122 unclamped) – not a near-miss, but a
 flat null result at substantially higher statistical power. More than doubling ($n = 3 \rightarrow 8$)
 the per-angle realizations therefore does *not* resolve Fix C’s failure at $\alpha = 85^\circ$: the
 data-scarcity explanation, while a reasonable hypothesis from the diagnostics above, is
 not supported once tested directly at the scale practically available to this project. The
 bottleneck – real seed-specific signal is demonstrably present in the observed input (9–
 15% pixel-level variation across seeds) yet is not exploited by the trained encoder at any
 tested sample size or FiLM-damping setting – should be reported as a genuinely unre-
 solved architecture/objective limitation rather than attributed to a data-volume shortfall
 that a modest seed increase was expected to fix. We report this as a strengthened neg-
 ative result: it directly pre-empts, rather than merely anticipates, the natural reviewer
 objection that more seeds would resolve the non-identifiability. (The 56 cross-realization
 pairs are clustered – eight query-realizations with seven sources each, and predictions vary
 only weakly across sources for a fixed query, which is itself the observed inertness of the
 conditioning pathway; this clustering, if anything, biases the test toward detecting a dif-
 ference, so the null result is conservative.) Table 6 summarises all six causal-substitution
 permutation tests run for Fix C across both FiLM-damping variants, both tested angles,
 and both sample sizes: none reaches significance, and the $n = 3 \rightarrow n = 8$ extension at
 $\alpha = 85^\circ$ leaves the same- and cross-realization correlation means essentially unchanged
 while increasing the cross-realization sample by nearly an order of magnitude.

Table 6: Six independent tests, two FiLM-conditioning strengths, two sample sizes – and not one shows a same- versus cross-realization separation a working causal mechanism would produce. Fix C causal-substitution permutation test results (Mann–Whitney U on same- versus cross-realization vertical-profile Pearson correlation). n : independent seeds at that angle; “pairs”: same-realization / cross-realization comparison counts. Unclamped $n=3$ same-/cross-realization means were not logged separately (only individual example values and the p -value were recorded); the text (Section 4.6) reports both populations in the ≈ 0.05 – 0.07 range, consistent with the damped $n = 3$ values shown here.

Variant	α ($^\circ$)	n	pairs (same/cross)	same- r	cross- r	p
Damped	80	3	3/6	0.073	0.073	0.36
Damped	85	3	3/6	0.055	0.055	0.55
Unclamped	80	3	3/6	≈ 0.05 – 0.07		0.27
Unclamped	85	3	3/6	≈ 0.05 – 0.07		0.73
Damped	85	8	8/56	0.127	0.126	0.35
Unclamped	85	8	8/56	0.124	0.122	0.53

4.7 Architecture-independent cross-validation: classical reconstruction and global connectivity agree with the diagnosis

Every remedy tested above (Fix A/B/C, and the seed-isolation ablation) is a point-wise, coordinate-conditioned neural model. A reasonable objection is that the failure is specific to that model family – some other, non-neural approach might recover the missing structure. We tested this directly with two architecture-independent checks that use no coordinate-conditioned neural network at all: a classical stochastic-reconstruction baseline, and an independently-computed global connectivity descriptor from a separate simulation campaign. Both converge on the same conclusion as Fix C.

A classical stochastic-reconstruction baseline. Two-point correlation functions do not, in general, uniquely determine a microstructure realization [Yeong and Torquato, 1998] – but the size of that gap is an empirical question for any specific material system, not a foregone conclusion. We measured it directly for these films using the Yeong–Torquato swap-based simulated-annealing algorithm: for each α , every z -slice of the true occupancy volume is reconstructed from scratch, starting from a random binary field at the correct volume fraction ϕ and repeatedly swapping an occupied/unoccupied voxel pair whenever the swap reduces the squared error between the candidate’s radially-

averaged two-point correlation $S_2(r)$ (periodic boundary, $r = 1\text{--}12$ voxels, short range $r \leq 3$ weighted $4\times$) and the true slice's $S_2(r)$, at fixed temperature schedule ($T_0 = 0.004$, geometric cooling, 18,000 swaps per slice). Because every accepted move is a $1 \leftrightarrow 0$ swap, ϕ is matched exactly by construction; only $S_2(r)$ is optimized. Reconstructed slices are then scored against the true slice with the same IoU metric used throughout Table 5.

Scope caveat, stated explicitly: this is a different task from the one Fix A/B/C were scored on. Fix A/B/C fit one coordinate function $f(x, y, z, \alpha) \rightarrow$ occupancy jointly across all nine α , evaluated on held-out voxel *samples* pooled across angles. The reconstruction here instead asks a complementary, arguably more information-scarce question: with *no* cross-angle training and *no* coordinate signal at all – only the volume fraction and two-point correlation of one realization, measured on that same realization – how much of it can be recovered? It is an information-theoretic ceiling test, not a drop-in fourth entry in Table 5.

Table 7 gives the result at all nine angles (25 of 151 z -slices sampled per angle, seed 0). At every non-degenerate angle, the swap-based search closes 70–96% of the initial random-field S_2 -matching error – i.e., it finds a configuration whose two-point correlation function is very close to the true one – yet IoU against the true realization remains low and falls monotonically as ϕ decreases. $\alpha = 60^\circ$ is a degenerate near-empty case ($\phi \approx 0.002$) included for completeness. **A classical algorithm operating directly on nearly-exact second-order statistics, with no architecture to blame, cannot recover the realization either.** This rules out a specific class of counter-explanation for the neural failure (that the tested architectures were simply the wrong ones) without weakening the diagnosis: the missing information is not resident in any second-order statistic of the film, for any method tested.

Cross-referencing an independently-computed global connectivity descriptor.

A separate simulation campaign (27 runs: all nine $\alpha \times$ three seeds, frozen periodic connected-component analysis, 0.5 nm voxels, film band $z \in [10\text{nm}, 0.90H]$) computed the largest connected-component fraction (LF) – the coarsest possible topology descriptor – independently of both P2's NIF work and the reconstruction above. Table 8 reports,

Table 7: Yeong–Torquato classical stochastic-reconstruction baseline, all nine trained angles (seed 0, 25/151 z -slices sampled per angle). S_2 error closure: $1 - (\text{final squared } S_2 \text{ error} / \text{initial random-field squared } S_2 \text{ error})$, i.e. how much of the gap to a perfect two-point-correlation match was closed by the swap-based search. IoU computed against ground truth on the same sampled slices. $\alpha = 60^\circ$ is near-empty ($\phi \approx 0.002$) and included only for completeness.

α ($^\circ$)	ϕ (occupancy)	S_2 error closure	IoU
60	0.000	– (degenerate)	1.000
65	0.612	93%	0.468
70	0.486	95%	0.338
72	0.458	96%	0.317
75	0.278	94%	0.172
80	0.221	92%	0.132
85	0.121	85%	0.068
87	0.072	70%	0.046
89	0.011	– (already near target)	0.007

for each α , the seed-to-seed spread of LF and of volume occupancy from that campaign, alongside whether Fix C tested that angle directly.

Table 8: Global connectivity (largest-fraction, LF) and occupancy, mean and seed-to-seed spread (max–min over 3 seeds), from an independently-run 27-run topology sweep. LF spread is 0.00000 up to $\alpha = 80^\circ$ and reaches only 0.013 at the most stochastic angle tested (89°) – versus the 9–15% *pixel-level* seed disagreement Fix C measured directly on the observed region at $\alpha = 85^\circ$ (Section 4.6).

α ($^\circ$)	occupancy (mean, spread)	LF (mean, spread)	Fix C tested?
60–75	0.63–0.83, ≤ 0.003	1.0000, 0.00000	no
80	0.500, 0.0024	1.0000, 0.00000	yes – saturated, no signal
85	0.304, 0.0032	0.9999, 0.00015	yes – real pixel-level signal, unexploited
87	0.198, 0.0018	0.9969, 0.00262	no
89	0.086, 0.0062	0.9822, 0.01306	no

Global connectivity is essentially seed-invariant at every angle tested – literally deterministic to five decimal places up to $\alpha = 80^\circ$, and still within 1.3 percentage points at 89° – a roughly 60–100 $\times$ gap relative to the 9–15% pixel-level seed disagreement Fix C independently measured at $\alpha = 85^\circ$.

A fourth check probes the same question one architecture class earlier and more cheaply than a full training run: does giving a model explicit spatial structure – rather than richer coordinate encoding, a different loss, or partial-realization conditioning with an MLP head – change the outcome? A small 3D U-Net (3 levels, 16/32/64 channels)

was built to take the coordinate-conditioned inputs used throughout this paper (voxel height and α , broadcast constant over x, y within each height, so that no ground-truth occupancy is ever fed to the model) and predict full-volume occupancy patches. Before any training-set evaluation, this architecture must pass a standard sanity gate: it should be able to drive the loss to ≈ 0 by memorizing a single fixed 32^3 patch. It could not. Three loss variants – focal loss at the paper’s default weighting, focal loss re-weighted for the patch’s 13% positive-voxel fraction, and plain binary cross-entropy – all plateaued at the same IoU (0.47–0.56), independent of the loss function used. An ablation isolates the cause: adding two explicit x, y coordinate channels (not a viable input for real training, since it would let the model memorize absolute position rather than generalize across patches) raises the memorization IoU to 1.000 in the same protocol. The mechanism is a direct consequence of the input design, not an optimization or capacity failure: height- and angle-only conditioning is constant over x, y at fixed height, and a translation-equivariant convolutional network maps a spatially-constant input field to a spatially-constant output field (up to boundary effects), so it cannot represent an x, y -varying occupancy texture from this input by construction. Spatial inductive bias alone therefore does not remedy the non-identifiability documented above – a CNN is subject to the same ceiling as the point-wise architectures of Table 5 whenever it is given the same coordinate-only information, and giving it more (absolute position) would trade the diagnosis for label leakage rather than resolve it.

A related but distinct question is whether Fix C’s own encoder – which *does* have access to real, per-realization structural information (the observed near-substrate region) rather than coordinates alone – was limited by its specific implementation rather than by the underlying task. Inspecting that encoder shows it ends in a global-average-pooling layer, collapsing the entire observed-region footprint into one vector before the FiLM-modulated decoder ever sees it – a bottleneck that discards exactly the per-column spatial information the causal-substitution diagnosis in Section 4.6 showed is present at $\alpha = 85^\circ$ (9–15% pixel-level variation across seeds). We therefore built a second variant that avoids this specific bottleneck: a resolution-preserving 2D encoder over the observed map, with

each query point’s prediction pixel-aligned to its own (x, y) location in that feature map via bilinear sampling (local, not global, conditioning), replacing FiLM modulation with direct concatenation. Before any full-scale training, this architecture was required to pass the same class of sanity check as the coordinate-only CNN above – memorize a small subset of one realization’s own points – and it did, reaching perfect accuracy in under 150 steps, confirming the pathway can carry and exploit local information when a target is directly learnable from it. Trained on the full partial-conditioning dataset and evaluated with the identical causal-substitution permutation test used for the original Fix C, the result is unchanged: same- and cross-realization vertical-profile correlations remain statistically indistinguishable at both tested angles (Mann–Whitney $p = 0.64$ at $\alpha = 80^\circ$, $p = 0.49$ at $\alpha = 85^\circ$ – if anything weaker separation than the original FiLM-based Fix C’s $p = 0.36/p = 0.55$), with same-realization predictions differing from cross-realization substitutions by no more than ordinary run-to-run noise (e.g. 0.339 same vs. 0.328–0.340 across all seven substituted sources, for a representative $\alpha = 85^\circ$ query). This rules out the global-pooling bottleneck as *the* explanation for Fix C’s failure: even a properly spatially-resolved, sanity-checked conditioning pathway is not exploited once trained at scale, strengthening rather than weakening the case that the observed-to-target correlation this task requires is either too weak to detect at the realization counts available or absent altogether in this class of stochastically-nucleated film, independent of which of the encoder architectures tested here is used to try to extract it.

Synthesis. Five independent methods, on five independent analyses of these films – Fix C’s causal-substitution permutation test (both its original FiLM-based encoder and the pixel-aligned local-conditioning variant above), the 27-run topology sweep, the Yeong–Torquato reconstruction above, and the coordinate-ablation check on a spatially-aware architecture – converge on one conclusion: seed-to-seed stochasticity in these films is real, substantial at the pixel level, and essentially invisible to every low-order descriptor tested (connectivity class, volume fraction, two-point correlation), regardless of whether that descriptor is consumed by a point-wise neural network, a spatially-convolutional one, a classical optimizer, or read directly off an independently-computed statistic. The negative

result in Table 5 is therefore not attributable to the specific architectures tried in this paper; it reflects where the recoverable information genuinely is (and is not) in this class of stochastically-nucleated GLAD film.

4.8 Limitations

Beyond the label-quality and multi-seed confounds already discussed above, the following scope limitations should be kept in mind when reading the results of this paper alongside other project documents.

1. *Single film height and pitch.* All results are for $h = 300$ nm, $\text{pitch} = 150$ nm; generalization to other pitches or heights is not tested.
2. *Ballistic model only.* The density fields are from a diffusion-free bead simulator; real Cu GLAD films at $\alpha > 80^\circ$ may have structural differences.
3. *Multi-seed label collision at $\alpha = 80^\circ, 85^\circ$.* Discussed in Section 4.5.
4. *Hard-label threshold.* The 0.5 threshold for binarisation is physically motivated (majority occupation) but arbitrary; sensitivity to threshold choice is not studied here.
5. *Occupied-fraction values are z -range-dependent and not directly comparable across this project’s own documents.* Table 3’s $\bar{y}_\alpha$ column uses the full $z \in [0, 151)$ voxel range. A separate, later analysis restricted to training slices $z \in [0, 121]$ reports different occupied fractions for the nominally same configurations (e.g. at $\alpha = 85^\circ$: 0.126 vs. 0.177 here; at $\alpha = 89^\circ$: 0.012 vs. 0.030 here) — a z -range definitional difference, not a data inconsistency, but one that was not disclosed here and should not be conflated across documents without noting which z -range each value uses.

5 Conclusion

We have investigated the use of a FILM-conditioned neural implicit field for scalar porosity and binary voxel occupancy prediction in ballistic helical GLAD simulations, with the

aim of providing an experimentalist with a fast surrogate for process-window screening.

We close each of the three objectives stated in Section 1, in the same order.

- **Objective 1 (scalar porosity surrogate) — answered: full-field density reconstruction within $[75^\circ, 87^\circ]$ is positive, but not a wider-domain result.**

A separately-configured NIF v5 variant achieves mean spatial Pearson $r = 0.984$ reconstructing the 3D density field at its 7 trained angles $\alpha \in \{75, 77.5, 80, 82.5, 85, 86, 87\}^\circ$.

This confirms the architecture family’s suitability as a density-level surrogate *within that window*; it does not establish, and this experiment does not test, generalisation to the wider $\{60, \dots, 89\}^\circ$ range used in Experiments 2–3, nor is it the same network as the shared architecture used there. A held-out test at three angles inside the trained window but withheld from training ($78.75^\circ, 83.5^\circ, 84.25^\circ$; Section 3.1.1) finds this correlation collapses to $r = 0.003$ mean, showing the $r = 0.984$ figure reflects per-angle recall rather than a genuinely angle-continuous learned field – the model does not interpolate even within its own trained window.

- **Objective 2 (voxel-level occupancy) — answered: soft-label spatial occupancy is negative.** BCE with max-normalised density labels freezes at loss = 0.541; all deposition angles remain trivial predictors, i.e. the connectivity/percolation-relevant voxel-level information motivating Objective 2 is not recovered by this labelling scheme. The physics-informed mass-conservation penalty corrects the per- α density bias but cannot restore binary spatial discrimination from soft labels: it is a density-calibration tool only, not a fix for the spatial-occupancy problem.

- **Objective 3 (root cause and remedy) — answered: diagnosed root cause; three candidate remedies tested, none sufficient at the current data scale, with the reasons for each failure diagnosed rather than left open.** Root cause: 90–98% of soft labels lie in $(0.1, 0.9)$, eliminating binary boundary gradients from the BCE gradient; an (α, z) -oracle diagnostic rules out architecture capacity and depth-signal absence as alternative explanations. Retraining with hard binary labels ($\{0, 1\}$, threshold 0.5) reduces BCE from 0.669 to 0.592 and achieves non-

trivial IoU for $\alpha = 89^\circ$ ($\Delta\text{IoU} = +0.029$); 6/7 other angles remain trivial due to two
 identified confounds: multi-seed label conflicts at $\alpha = 80^\circ$ and 85° , and balanced-
 sampling distortion of the test class distribution. Hard binary labels are therefore
 a *necessary but not sufficient* condition. A direct follow-up combining seed-isolated
 splits and focal loss (early-stopped at epoch 291) was subsequently run and did *not*
 resolve spatial learning: 5/7 angles remained statistically indistinguishable from the
 trivial predictor ($\Delta\text{IoU} < 0.0001$ at $\alpha = 65\text{--}85^\circ$), with only modest improvement at
 $\alpha = 87^\circ$ ($\Delta\text{IoU} = 0.005$) and $\alpha = 89^\circ$ ($\Delta\text{IoU} = 0.058$); the aggregate IoU (0.419)
 is not representative of per-angle performance. This rules out multi-seed pooling as
 the dominant confound on its own. Three further remedies were then tested directly
 (Section 4.6): a multi-resolution hash-grid spatial encoder (higher-capacity local en-
 coding, not a structural inductive bias), continuous-density regression (a loss/target
 reframing), and partial-realization conditioning (a task reframing, conditioning pre-
 diction of a realization’s upper region on an encoded latent of that same realization’s
 own near-substrate region). The first two produce no meaningful improvement over
 the hard-label baseline at any angle. The third fails a 3-seed causal-substitution
 permutation test at both multi-seed angles, for two mechanistically distinct rea-
 sons: at $\alpha = 80^\circ$ the near-substrate observed region is itself $> 99.8\%$ saturated
 regardless of seed, leaving nothing informative to condition on; at $\alpha = 85^\circ$ real seed-
 distinguishing signal exists in the observed region (13–15% pairwise pixel variation)
 but the trained encoder’s influence on the output remains $\sim 10^{-3}$ in magnitude.
 A direct follow-up retrained the conditioning pathway fully unclamped ($10\times$ larger
 allowed influence, removing the 10% FiLM damping inherited from the baseline ar-
 chitecture) to test whether the damping itself was responsible: the permutation test
 still fails at both angles with the same order-of-magnitude same/cross difference,
 ruling out conditioning-pathway capacity as the bottleneck and narrowing it, provi-
 sionally, to the number of realizations sharing each angle (three). This data-scarcity
 hypothesis was then tested directly: five additional simulations at $\alpha = 85^\circ$ (seeds 3–
 7) brought that angle to $n = 8$ realizations, and both Fix C variants were retrained

from scratch on the enlarged dataset. **The permutation test still fails at $n = 8$** for both variants (Mann–Whitney $p = 0.35$ damped, $p = 0.53$ unclamped, evaluated over 56 cross-realization pairs versus 6 previously), with same- and cross-realization correlation means remaining statistically indistinguishable (0.127 vs. 0.126; 0.124 vs. 0.122) rather than merely falling short of significance. More than doubling the per-angle realizations therefore does *not* resolve Fix C’s failure. A higher-capacity local spatial encoding, loss/target reframing, and now a practically-achievable increase in per-angle realizations are all ruled out as sufficient fixes; partial-realization conditioning’s failure at $\alpha = 85^\circ$ – where real seed-specific signal is demonstrably present in the input yet is not exploited by the trained encoder at any tested sample size or FiLM-damping setting – is reported as a genuine, tested negative result rather than an artefact of insufficient data, pending architectural or objective-function changes not explored here.

This diagnosis is further strengthened by four architecture-independent cross-checks (Section 4.7) that use no coordinate-conditioned neural network at all: a classical Yeong–Torquato stochastic-reconstruction baseline, operating directly on the two-point correlation function of each realization, closes 70–96% of the initial random-field correlation-matching error yet still recovers only 7–47% intersection-over-union with the true realization at non-degenerate angles; an independently-run 27-configuration topology sweep shows the global largest-connected-fraction descriptor is seed-invariant to five decimal places up to $\alpha = 80^\circ$ and within 1.3 percentage points at $\alpha = 89^\circ$, a 60–100 $\times$ gap relative to the 9–15% pixel-level seed disagreement measured directly; a coordinate-conditioned convolutional network is shown, by construction, to be representationally incapable of producing an x, y -varying occupancy texture from height/angle-only input; and a spatially-resolved, sanity-checked local-conditioning variant of the partial-realization encoder – built specifically to rule out the global-average-pooling bottleneck of the original encoder – still shows no learned signal once trained at scale. Five independent methods, on five independent analyses of these films, therefore converge on one conclusion: seed-to-seed stochasticity in these films is real, substantial at the pixel level, and essentially invisible

to every low-order descriptor tested, regardless of whether that descriptor is consumed by a point-wise neural network, a spatially-convolutional one, a classical optimizer, or read directly off an independently-computed statistic. The negative result is therefore not attributable to the specific architectures tried in this paper; it reflects where the recoverable information genuinely is (and is not) in this class of stochastically-nucleated GLAD film.

These results establish a stepwise label-quality diagnosis framework — soft-zone fraction analysis, oracle predictor test, and hard-label ablation — applicable to any simulation-based NIF training pipeline where continuous density fields are the natural simulation output but binary voxel structure is the prediction target. For an experimentalist, the practical takeaway is that a fast scalar-porosity surrogate is already usable within its trained angular window, while a voxel-resolved, connectivity-aware surrogate is not yet available: a higher-capacity local spatial encoding alone does not supply it (Section 4.6), and the most promising remaining direction, partial-realization conditioning, requires more per-angle realizations and a stronger conditioning pathway than were available in this study before it can be properly evaluated.

Author contributions (CRediT)

O.B.H.A.: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. F.C.A.: Supervision, Writing – review & editing. B.G.: Supervision, Writing – review & editing. N.M.: Supervision, Writing – review & editing.

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Data availability

Simulation data, voxel files, and training scripts are available from the authors upon reasonable request.

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